

Lesson 16: Limit Comparison Test

Assessment Rubrics

- **Requirement 1: Positive Terms (5%):** Both series $\sum a_n$ and $\sum b_n$ must have positive terms ($a_n > 0, b_n > 0$) for all n (or at least for all $n > N$ for some integer N).
- **Requirement 2: Selection of Comparison Series (b_n)(5%):** Identify the dominant terms in the numerator and denominator of a_n to determine the behavior for large n . The comparison series b_n is typically a simple p -series ($\sum \frac{1}{n^p}$) or geometric series ($\sum ar^n$) whose behavior is already known.
- **Requirement 3: Calculation of the Limit (L) (70%):** Calculate the limit $L = \lim_{n \rightarrow \infty} \frac{a_n}{b_n}$.
- **Requirement 4: (20%): Interpretation of L (The Three Cases):**
 - **Case 1: $0 < L < \infty$ (Finite and Positive):** Both series $\sum a_n$ and $\sum b_n$ either converge or diverge together.
 - **Case 2: $L = 0$:** If $\sum b_n$ converges, then $\sum a_n$ also converges.
 - **Case 3: $L = \infty$:** If $\sum b_n$ diverges, then $\sum a_n$ also diverges.